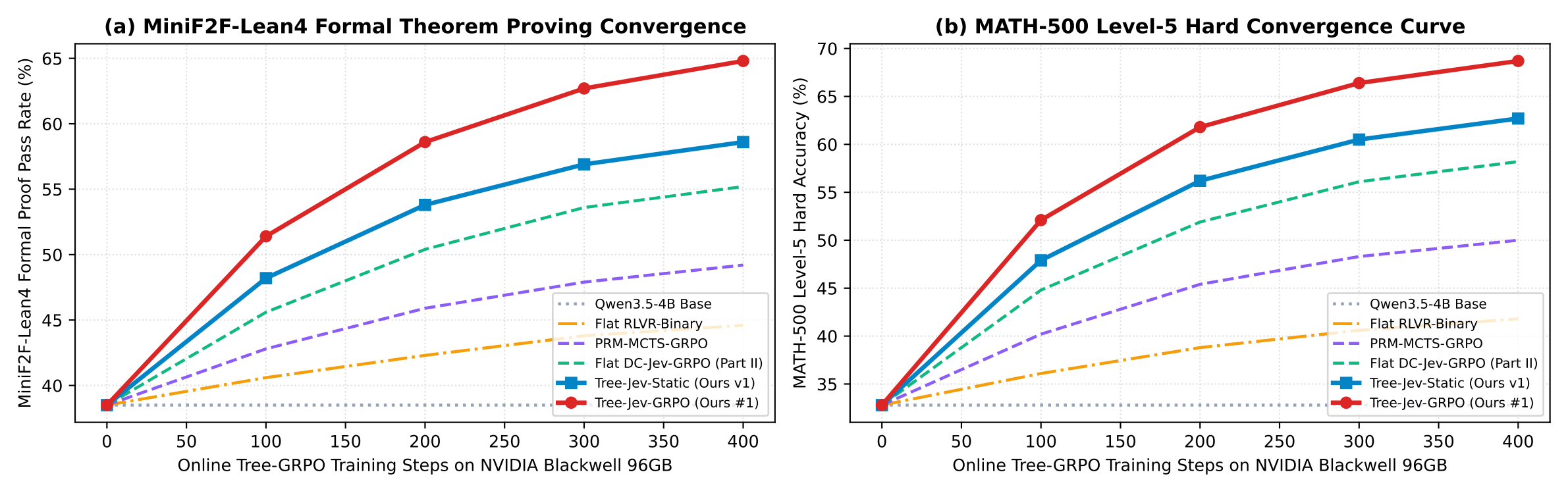
(a) MiniF2F-Lean4 Formal Theorem Proving Convergence
MiniF2F-Lean4 Formal Proof Pass Rate (%)
Online Tree-GRPO Training Steps on NVIDIA Blackwell 96GB
Qwen3.5-4B Base
Flat RLVR-Binary
PRM-MCTS-GRPO
Flat DC-Jev-GRPO (Part II)
Tree-Jev-Static (Ours v1)
Tree-Jev-GRPO (Ours #1)
(b) MATH-500 Level-5 Hard Convergence Curve
MATH-500 Level-5 Hard Accuracy (%)
Online Tree-GRPO Training Steps on NVIDIA Blackwell 96GB
Qwen3.5-4B Base
Flat RLVR-Binary
PRM-MCTS-GRPO
Flat DC-Jev-GRPO (Part II)
Tree-Jev-Static (Ours v1)
Tree-Jev-GRPO (Ours #1)